Plant Disease Detection System

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***Abstract*—Strawberry is a high-value, disease-susceptible crop for which timely and accurate disease identification is essential to limit yield loss and reduce unnecessary pesticide use. Over the past several years, deep-learning-based computer vision has progressively replaced manual visual inspection as the preferred approach for strawberry disease monitoring, moving from two-stage cascade detectors with domain-pretrained backbones to open-set recognition schemes and, most recently, to a family of lightweight, real-time YOLO-based architectures suitable for edge deployment. This paper presents a structured review of five representative studies that collectively span this evolution: (1) a PlantNet-based cascade Faster R-CNN detector, (2) a deep-metric-learning post-filter for known/unknown disease discrimination, (3) an attention- and sampling-augmented YOLOv8 framework, (4) a lightweight YOLO11-Strawberry model built around a VoV-Faster block and a multi-scale feature channel fusion attention (MSFCA) module, and (5) D-YOLO, a MobileNetv3-based YOLOv8 variant employing BiFPN, Contextual Transformer (CoT) attention, and Wise-IoU loss. We summarize the datasets, network designs, loss functions, and evaluation protocols used in each study, present a comparative analysis of reported precision, recall, and mAP against model size and computational cost, and discuss recurring challenges including class imbalance, background interference, small-target detection, multi-scale variation, and the transition from cloud-based to edge inference. We conclude by identifying open research directions, including unified open-set detection, cross-domain generalization, and standardized benchmark datasets for strawberry disease detection.**

***Index Terms*—strawberry disease detection, deep learning, object detection, YOLO, Faster R-CNN, deep metric learning, attention mechanism, lightweight network, precision agriculture**

I. Introduction

Strawberry (*Fragaria* × *ananassa*) is a high-value fruit crop that is highly vulnerable to fungal and bacterial diseases due to its thin skin, high moisture content, and dense canopy structure [1], [5]. Diseases such as anthracnose fruit rot, gray mold, powdery mildew, angular leafspot, and leaf blight can spread rapidly within greenhouses, causing severe yield and quality losses if not detected early [1], [2]. Traditional disease diagnosis relies on manual visual inspection by farmers or laboratory analysis by experts, both of which are labor-intensive, subjective, slow, and prone to error at scale [**?**], [4].

The emergence of deep convolutional neural networks (CNNs) has enabled automated, vision-based disease detection systems that promise faster, more consistent, and more scalable monitoring. Two broad families of approaches have been explored: (i) *classification* approaches, in which a human first isolates a symptomatic region and a CNN classifies the disease, and (ii) *detection* approaches, in which an object detector jointly localizes and classifies diseased regions directly from unconstrained field or greenhouse images [1]. Detection-based pipelines are generally preferred for automated and robotic monitoring because they impose fewer constraints on image acquisition and can process video streams.

This review synthesizes five recent studies that represent distinct stages in the evolution of strawberry disease detection research:

- **Kim et al. (2021)** [1] — a two-stage cascade Faster R-CNN detector using a PlantNet backbone pretrained on the PlantCLEF/LifeCLEF 2017 dataset, designed to mimic the coarse-to-fine visual inspection strategy of human experts.
- **You et al. (2022)** [2] — a Faster R-CNN detector coupled with a deep metric learning (DML)-based post-filter and K-nearest neighbor (K-NN) classifier that can flag previously unseen ("unknown") disease-like symptoms, addressing open-set recognition.
- **Xie et al. (2024)** [3] — a YOLOv8-based framework enhanced with three alternative modules (CBAM, DySample, ODConv), evaluated with K-fold cross-validation and accelerated using TensorRT for real-time cloud/edge deployment.
- **Xie et al. (2025)** [4] — YOLO11-Strawberry, a lightweight variant of YOLO11s incorporating a VoV-Faster block (FasterNet + VoV-GSCSP) and a Multi-Scale Feature Channel Fusion Attention (MSFCA) module, targeting edge-device deployment.
- **Wu et al. (2025)** [5] — D-YOLO, a YOLOv8s variant using a MobileNetv3 backbone, a Bidirectional Feature Pyramid Network (BiFPN), Contextual Transformer (CoT) attention, and Wise-IoU (W-IoU) loss, evaluated on both strawberry and maize disease datasets for cross-domain generalization.

The remainder of this paper is organized as follows. Section II reviews the datasets used across the five studies. Section III describes the network architectures and key technical contributions of each work. Section IV presents a comparative analysis of reported performance and computational efficiency. Section V discusses common challenges and design trends. Section VI concludes the review and outlines future research directions.

II. Datasets and Disease Categories

A summary of the datasets employed in the reviewed studies is given in Table I. Dataset scale ranges from under a thousand images [5] to nearly eight thousand images [3], and the number of disease categories ranges from four to nine, reflecting differing scopes (pure disease detection vs. joint development-stage and health-state detection).

Several observations follow from Table I. First, the absence of a single, standardized, large-scale public strawberry disease detection benchmark forces every group to construct its own dataset, which severely limits direct cross-study comparability of reported accuracy figures. Second, class imbalance is a pervasive issue: normal/healthy samples typically dominate (e.g., 2,000 of 4,725 original images in [1] are "normal"; "healthy" is similarly the majority class in [5]), while rare diseases such as anthracnose fruit rot or powdery mildew fruit are chronically under-represented. Third, most datasets are collected under uncontrolled field or greenhouse lighting, deliberately avoiding artificial standardization, in order to preserve the realistic appearance of complex backgrounds, occlusion, and illumination variance that a deployed system must tolerate [3], [4].

III. Architectures and Technical Contributions

A. Cascade Detection with Domain-Pretrained Backbones (Kim et al., 2021)

Kim et al. [1] propose a two-stage cascade detector that imitates the human expert workflow of first identifying a suspicious region and then scrutinizing it closely. Both stages share a ResNet152 backbone augmented with a Feature Pyramid Network (FPN), but critically, the backbone is pre-trained on the PlantCLEF dataset (LifeCLEF 2017 challenge, $\sim$1.25M filtered plant images across 10,000 categories) rather than on ImageNet. This backbone, termed *PlantNet*, is argued to encode more relevant low-level domain knowledge (leaf/flower/fruit textures) than a generic ImageNet-pretrained network. In Stage 1, a low-confidence-threshold detector flags any suspicious region as "normal," "abnormal," or "background object"; adjacent abnormal boxes are merged, background objects are excluded, and the merged region is cropped and resized for Stage 2, which performs fine-grained classification into seven disease categories using an identical ResNet152+FPN architecture with a higher IoU threshold (0.5) to boost precision.

Two key ablations support the design: (i) replacing an ImageNet-pretrained backbone with PlantNet raises mean Average Precision (mAP) from 83.13% to 86.40% in a single-stage detector, and (ii) adding the cascade structure raises mAP further to 91.65% (a 5.25-point gain over the single-stage PlantNet detector), at the cost of increased inference time (0.241 s vs. 0.662 s per image).

B. Open-Set Recognition via Deep Metric Learning (You et al., 2022)

You et al. [2] extend the cascade-detection paradigm to address a critical practical limitation: object detectors trained only on known disease classes cannot reliably flag genuinely novel or ambiguous symptoms, often misclassifying them with high confidence. Their pipeline couples an FPN-based Faster R-CNN (first stage) with a deep metric learning (DML) post-filter (second stage) trained using a ResNet50 backbone with combined margin triplet loss and cross-entropy loss. The triplet loss embeds 256-dimensional features such that same-class samples cluster tightly while different classes are pushed apart, deliberately leaving "room" in the embedding space for unknown symptoms.

Two classifiers operate on the same embedding: a softmax classifier restricted to 12 known categories (8 diseases + 4 normal parts), and a K-nearest-neighbor (K-NN, $K = 13$) classifier that additionally recognizes an "unknown" class using known-unknown reference samples never seen during backbone/softmax training. Final decisions for known classes combine both classifiers' probabilities multiplicatively (Eq. 6 in the original paper), while unknown detections rely solely on the K-NN output. Empirically, adding the DML post-filter raises detection mAP from 90.6% (Faster R-CNN alone) to 92.8% (combined decision), and the separate K-NN-based ROI classifier achieves 97.8% accuracy across known, normal, and unknown categories. Field testing on genuinely unseen disease types (e.g., powdery mildew runner, never used in training) shows that most such cases are correctly routed to the "unknown" category rather than being confidently misclassified as a known disease — a crucial safety property for real deployment.

C. Attention- and Sampling-Enhanced YOLOv8 (Xie et al., 2024)

Xie et al. [3] depart from the two-stage R-CNN paradigm in favor of the single-stage YOLOv8 detector, motivated by the need for real-time inference. Three architectural variants are compared, each modifying the YOLOv8 head/neck:

- **YOLOv8+CBAM**: inserts a Convolutional Block Attention Module (channel attention via shared MLP over max/avg-pooled features, followed by spatial attention via a convolution over pooled channel maps) after the final C2f block in the head.
- **YOLOv8+DySample**: replaces static nearest-neighbor upsampling with a Super-Lightweight Dynamic Upsampling Operator that performs point-wise differential sampling, reducing computation relative to learned deconvolution while improving upsampling fidelity.
- **YOLOv8+ODConv**: replaces standard convolutions in the C2f/bottleneck modules with Omni-Dimensional Dynamic Convolution, which learns complementary attention across all four kernel dimensions (spatial, input channel, output channel, and kernel number) via parallel sigmoid-gated branches.

Using K-fold cross-validation on a self-collected 7,980-image, 7-class dataset, YOLOv8+DySample achieves the best F1 score (97.086%) while YOLOv8+CBAM achieves the highest mAP@0.5 (98.257%) and the most stable/smooth training loss curves. Notably, ODConv *underperforms* the vanilla YOLOv8

TABLE I
SUMMARY OF DATASETS USED ACROSS THE REVIEWED STUDIES

Study	Total Images	Acquisition	Categories	Notes
Kim et al. [1]	4,735 (4,560 own + 175 Forestry Images)	Mobile phones, greenhouses; public Forestry Images archive	7 disease classes + normal + background (angular leafspot, anthracnose fruit rot, gray mold, leaf blight, leaf spot, powdery mildew fruit/leaf)	Backbone pretrained separately on PlantCLEF (LifeCLEF 2017, ∼1.45M raw / 250K clean images)
You et al. [2]	7,230	Cellular phones, multiple greenhouses	8 known diseases + 4 normal parts + unknown class	Known-unknown samples deliberately included to train K-NN unknown detector
Xie et al. (2024) [3]	7,980	Field photography	7 categories (angular leafspot, anthracnose fruit rot, blossom blight, gray mold, leaf spot, powdery mildew fruit/leaf)	No public strawberry pest/disease dataset existed; original data collected and labeled with LabelImg
Xie et al. (2025) [4]	900	Web crawling + smartphone photography	5 classes (Phytophthora fragariae, anthracnose fruit rot, gray mold, powdery mildew, healthy fruit)	700/100/100 train/val/test split; bounding-box size analysis shows predominance of small targets
Wu et al. [5]	1,301 (after augmentation)	Public AI Challenger dataset + own greenhouse photography + Douyin (TikTok) user videos	3 fruit stages (flower, fruit, ripe) × 5 health states (healthy, fertilizer-deficient, calcium-deficient, powdery mildew, greyleaf)	Heavy class imbalance addressed via targeted augmentation and dynamic train/val/test split

baseline (95.181% F1 vs. 96.013%), illustrating that not every attention/dynamic-convolution mechanism transfers well to this domain. Finally, TensorRT acceleration (FP16/INT8 quantization) reduces per-image inference time roughly threefold (e.g., from ∼4–5 ms to ∼0.6 ms), and the resulting model is deployed as a cloud API with a mobile client.

D. Lightweight YOLO11 with Multi-Scale Attention (Xie et al., 2025)

Xie et al. [4] target edge deployment explicitly, building on YOLO11s (chosen over YOLO11n/m for its speed–accuracy balance) and introducing two complementary modules:

- **VoV-Faster block**: combines FasterNet's Partial Convolution (PConv, which convolves only a subset of channels to reduce memory access and redundant computation) with the one-time-aggregation VoV-GSCSP structure, replacing the computationally heavier C3KA/C3KB modules in YOLO11's neck. This yields a 19% parameter reduction and 24.4% FLOPs reduction in isolation, with negligible accuracy loss.
- **Multi-Scale Feature Channel Fusion Attention (MSFCA)**: splits input features along the channel dimension into three parallel branches, applies scale-specific convolutions/combinations to each, recombines them, and applies a mean-pooling-based channel attention ("Scale") operation. This targets the specific challenge — confirmed by the paper's own bounding-box size analysis — that most strawberry disease regions occupy a small fraction of the full image, causing feature information loss through the network's depth. In isolation, MSFCA improves precision by 2.3 points, recall by 3.1 points, and mAP50 by 1.9 points over the YOLO11s baseline, and Grad-CAM-style heatmaps confirm markedly improved localization on disease regions vs. background.

Combining both modules (YOLO11-Strawberry) yields 93.2% precision, 92.5% recall, and 95.3% mAP50, while *simultaneously* reducing parameters by 10.6% (9.4M → 8.4M) and FLOPs by 17.4% (21.3G → 17.6G) relative to YOLO11s — a rare instance of joint accuracy and efficiency improvement rather than a trade-off. On a Jetson Xavier NX edge device, throughput increases from 21 FPS (YOLO11s) to 27 FPS (YOLO11-Strawberry), meeting real-time thresholds. Comparative experiments show YOLO11-Strawberry outperforming Faster R-CNN, SSD, YOLOv8s, YOLOv10s, and YOLO11n/m/s on the joint precision/recall/mAP/parameter/FLOPs criteria.

E. Cross-Domain Lightweight Detection: D-YOLO (Wu et al., 2025)

Wu et al. [5] present the most aggressively compressed architecture reviewed here, D-YOLO, built on YOLOv8s with four coordinated modifications:

1) **MobileNetv3 backbone**: replaces CSPDarknet with depthwise-separable convolutions and Squeeze-and-Excitation (SE) channel attention, cutting backbone parameters drastically (2.3M vs. 11.1M total model parameters for baseline YOLOv8s).
2) **Concat-BiFPN**: a bidirectional (top-down + bottom-up) feature pyramid with learnable per-edge fusion weights, replacing YOLOv8's PAN-Net neck to better balance contributions from features of different input resolutions.
3) **Contextual Transformer (CoT) attention**: three CoT blocks are inserted between the feature-fusion and detection heads; CoT fuses static context (via 3×3 convolution over keys) with dynamic multi-head attention (via 1×1 convolution over the query–key encoding), improving sensitivity to small, dense targets and long-range dependencies relative to conventional local self-attention.
4) **Wise-IoU (W-IoU) v3 loss**: replaces Complete IoU (CIoU) loss with a non-monotonic focusing-factor formulation that dynamically down-weights very easy or

very poor anchor boxes, improving robustness to the pronounced class imbalance in the dataset.

Backbone comparisons (Table 3 in [5]) show MobileNetv3 achieving the largest reduction in FLOPs (79.6%) and parameters (77.8%) among five lightweight alternatives (GhostNet, ShuffleNet, EfficientViT, MobileNeXt), at some cost to raw accuracy (89.0% vs. 91.4% mAP for full YOLOv8s). Ablation results show BiFPN, CoT, and W-IoU each contributing incremental mAP gains (to a final 89.6% training mAP / 90.5% test mAP), while keeping the parameter count near 3.1M and FLOPs near 7.1G — a 72.0% parameter and 75.1% FLOPs reduction relative to YOLOv8s, with total training time cut to 1.4 h (vs. 3.45 h for YOLOv8s and >35 h for Faster R-CNN/RetinaNet). Notably, D-YOLO is also validated on an unrelated maize disease dataset, where it achieves the *highest* mAP (94.5%) among all compared lightweight and full-scale models, demonstrating cross-domain generalization rather than task-specific overfitting.

IV. Comparative Analysis

Table II consolidates the headline detection metrics, parameter counts, and computational cost reported by each study. Because each group uses a different dataset, class taxonomy, and train/test split, these numbers should be interpreted as indicative of relative design trade-offs rather than as a strictly controlled benchmark.

Several trends emerge from this comparison:

- **Accuracy–efficiency trade-off is not fixed.** Earlier two-stage cascade designs [1], [2] achieve strong mAP (91–93%) but at a substantial computational cost (hundreds of milliseconds per image) unsuitable for edge deployment. The YOLO-family studies [3]–[5] progressively push inference toward real time, but interestingly, the most aggressively compressed model, D-YOLO [5], sacrifices roughly 1–5 mAP points relative to full-size YOLOv8s/YOLO11s, whereas YOLO11-Strawberry [4] achieves *both* higher accuracy and lower computational cost than its own baseline, indicating that attention/module design quality matters more than raw parameter count.
- **Attention mechanisms are not universally beneficial.** CBAM and CoT attention modules consistently improve detection metrics across studies [3]–[5], whereas OD-Conv underperforms the YOLOv8 baseline in [3]. This suggests that the specific placement (head vs. neck vs. backbone) and the nature of the attention (channel/spatial vs. dynamic convolutional kernels) must be tuned to the target domain rather than adopted uniformly.
- **Backbone domain pretraining remains valuable.** Kim et al.'s PlantNet backbone [1], pretrained on a plant-specific dataset (PlantCLEF) rather than ImageNet, yields a 3.2-point mAP improvement, echoed by You et al.'s continued use of the same PlantNet initialization [2]. None of the more recent YOLO-based studies [3]–[5] explore plant-domain pretraining, instead relying on standard ImageNet/COCO initialization — a potentially underexplored avenue for further gains.
- **Open-set recognition remains rare.** Only You et al. [2] explicitly addresses the detection of previously unseen disease symptoms, a capability of high practical importance given that real greenhouses will inevitably present novel or ambiguous symptoms. None of the more recent lightweight YOLO variants incorporate an analogous unknown-class mechanism, representing a gap between practical need and current lightweight-model research focus.
- **Small-object and background-interference handling is a recurring bottleneck.** Both Xie et al. (2025) [4] and Wu et al. [5] explicitly motivate their attention modules (MSFCA, CoT) by analyzing the small relative size of diseased regions and the complexity of natural backgrounds (leaves, soil, plastic mulch, other plants), and both report confusion primarily between diseased/healthy classes and background rather than between disease types — consistent with Kim et al.'s original design decision to explicitly model a "background object" class [1].

V. Discussion: Common Challenges and Emerging Trends

A. Class Imbalance

All five studies confront severe class imbalance, most commonly an overrepresentation of healthy/normal samples relative to rare diseases (e.g., anthracnose fruit rot, powdery mildew fruit). Mitigation strategies include targeted data augmentation restricted to minority classes [5], Online Hard Example Mining (OHEM) [1], Task-Aligned dynamic sample assignment [3], and loss reweighting via Wise-IoU's non-monotonic focusing factor [5]. Notably, Wu et al. observe that *excessive* augmentation can worsen imbalance by amplifying already-dominant visual patterns, underscoring that augmentation strategy — not merely augmentation volume — is critical.

B. Multi-Scale and Small-Target Detection

Both later studies [4], [5] perform explicit bounding-box size analysis and find that most diseased regions are small relative to the full image. Feature Pyramid Networks (used in all five studies in some form: FPN [1], [2], PANet/C2f in YOLOv8 [3], and BiFPN [5]) mitigate this via multi-scale feature aggregation, while attention modules (MSFCA [4], CoT [5]) further re-weight channels/positions to preserve small-object signal against being "washed out" by deeper layers.

C. From Cloud to Edge Deployment

A clear trajectory across the five studies is the shift from research-grade GPU inference (Titan Xp [1]; unspecified GPU [2]) toward explicit edge-deployment targets: TensorRT-accelerated cloud API with mobile client [3], Jetson Xavier NX benchmarking at 27 FPS [4], and D-YOLO's 2.3–3.1M-parameter footprint explicitly designed for resource-constrained devices [5]. This mirrors the broader precision-

TABLE II
COMPARATIVE SUMMARY OF REPORTED PERFORMANCE ACROSS REVIEWED STUDIES

Study	Best Configuration	Precision	Recall	mAP	Params	FLOPs / Speed
Kim et al. [1]	Cascade PlantNet-ResNet152 Faster R-CNN	—	—	91.65%	Not reported	0.662 s/image
You et al. [2]	Faster R-CNN + Softmax + K-NN (combined)	—	—	92.8% (known); 97.8% acc. (K-NN incl. unknown)	Not reported	Not reported
Xie et al. (2024) [3]	YOLOv8 + CBAM / + DySample	97.5% / 98.0%	96.2% / 96.2%	98.26% / 98.04% (mAP50)	Comparable to YOLOv8s ($\sim$11M)	$\sim$0.6 ms (TensorRT FP16)
Xie et al. (2025) [4]	YOLO11-Strawberry	93.2%	92.5%	95.3% (mAP50)	8.4M (-10.6%)	17.6G FLOPs (-17.4%); 27 FPS (Jetson Xavier NX)
Wu et al. [5]	D-YOLO	86.6% (test: 87.6%)	87.8% (test: 85.4%)	89.6% train / 90.5% test	3.1M (-72.0% vs. YOLOv8s)	7.1G FLOPs (-75.1%); 1.4 h train time

agriculture trend toward on-device, real-time monitoring rather than centralized post-hoc image analysis.

D. Lack of Standardized Benchmarks

No two studies reviewed here use the same dataset, class taxonomy, or evaluation protocol. This absence of a standardized, large-scale, publicly available strawberry disease benchmark (analogous to PlantVillage for classification) is arguably the single largest obstacle to rigorous comparative progress in this sub-field, and several papers explicitly note constructing their own dataset due to the lack of a suitable public alternative [1], [3].

VI. CONCLUSION AND FUTURE DIRECTIONS

This review has traced the evolution of strawberry disease detection from cascade Faster R-CNN architectures with domain-pretrained backbones [1], through open-set-aware deep metric learning post-filters [2], to a family of increasingly lightweight, attention-augmented YOLO variants optimized for real-time and edge inference [3]–[5]. Across this progression, three consistent themes emerge: the value of multi-scale feature fusion and attention for handling small, background-confusable disease regions; the necessity of addressing severe class imbalance through both data-level and loss-level interventions; and an accelerating shift toward lightweight architectures suitable for on-device, real-time deployment in greenhouses and fields.

Several directions merit further investigation. First, combining the open-set recognition capability demonstrated by You et al. [2] with the lightweight, real-time architectures of [4], [5] would yield a system that is both deployable on edge hardware and safely aware of its own detection limits — an important property currently missing from all lightweight variants reviewed. Second, plant-domain pretraining (as in PlantNet [1]) has not been revisited in the more recent YOLO-based literature and may offer further accuracy gains when combined with modern lightweight backbones. Third, the field would benefit substantially from a standardized, large-scale, multi-institution strawberry disease benchmark dataset with consistent class definitions, enabling fair, reproducible comparison across architectures — mirroring the role PlantVillage has played for leaf disease classification. Finally, cross-domain validation, as performed by Wu et al. on a maize dataset [5], should become standard practice to demonstrate that architectural improvements generalize beyond a single crop and dataset.

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